

20250313高代习题课

Throughout K is a number field

1. Let a linear space $P_n := \{f(x) \in K(x) \mid \deg f(x) < n\}$. Pick $a_1, \dots, a_n \in F$ such that $a_i \neq a_j$ for any $i \neq j$. Show that $f_j(x) := \prod_{i \neq j} (x - a_i) \quad (1 \leq j \leq n)$ form a basis of P_n .
2. Assume $K_1 \subseteq K_2 \subseteq K_3$ such that $\dim_{K_1} K_2 = m$, and $\dim_{K_2} K_3 = n$ (as linear spaces) show that $\dim_{K_1} K_3 = mn$.
3. V is a n -dimensional linear space on number field K , $V_1, V_2, \dots, V_s$ are s proper subspaces of V . Prove that
 - (1) $\exists \alpha \in V$, s.t. $\alpha \notin V_1 \cup V_2 \cup \dots \cup V_s$;
 - (2) $\exists$ a basis of V , $\varepsilon_1, \varepsilon_2, \dots, \varepsilon_n$, s.t. $\{\varepsilon_1, \dots, \varepsilon_n\} \cap (V_1 \cup \dots \cup V_s) = \emptyset$.
4. "Magic Square" is a n -order square matrix such that the sum of numbers on each row, column, diagonal, and secondary diagonal is equal.
Let M_n be the set of n -order magic squares on $\mathbb{R}$. Clearly, M_n is a linear space on $\mathbb{R}$. What is the dimension of M_3 ?

5. Assume real number series set $S = \{(x_1, x_2, \dots, x_n, \dots) \mid x_n = x_{n-1} + x_{n-2}, n \geq 3\}$,

and S forms a linear space on real number field $\mathbb{R}$ by

$$\text{addition: } (x_1, \dots, x_n, \dots) + (y_1, \dots, y_n, \dots) = (x_1 + y_1, \dots, x_n + y_n, \dots)$$

$$\text{scalar multiplication: } k(x_1, \dots, x_n, \dots) = (kx_1, \dots, kx_n, \dots).$$

(1) Find a basis of S , and calculate $\dim S$.

(2) Find a basis of S , consisting of some geometric series. (等比数列)

(3) Find the general term formula F_n of the Fibonacci series $(0, 1, 1, 2, 3, 5, 8, \dots)$.

6. (Quotient spaces). Let V be a linear space over K , and $U \subseteq V$ is a subspace.

For any $v \in V$, we define the U -coset (U -陪集) of v :

$$v + U := \{v + u \mid u \in U\}.$$

Let $S := \{v + U \mid v \in V\}$ and define

$$(v_1 + U) + (v_2 + U) = (v_1 + v_2) + U; \quad k \cdot (v_1 + U) = k v_1 + U. \quad \text{for any } v_1, v_2 \in V, \text{ and } k \in K.$$

Prove (1) Either $v_1 + U = v_2 + U$ or $(v_1 + U) \cap (v_2 + U) = \emptyset$

(2) $v_1 + U = v_2 + U$ iff $v_1 - v_2 \in U$.

(3) Both $+$ and $\cdot$ do not depend on the choice of v ,
i.e., if $v_1 + U = v_1' + U$, $v_2 + U = v_2' + U$, then

$$(v_1 + U) + (v_2 + U) = (v_1' + U) + (v_2' + U), \text{ and } k(v_1 + U) = k(v_1' + U)$$

(4) $(S, +, \cdot)$ is a linear space. (we say S is the quotient space, denoted by V/U).

7. Let V is a n -dimensional linear space on K . $U \subseteq V$ is a subspace.

W is the complementary space of U . Prove that.

(补空间)

$$\dim V/U = \dim V - \dim U, \text{ and } \exists \text{ linear isomorphism } \varphi: W \mapsto V/U$$

(线性同构)